WORKED SOLUTION

STEP I Specimen paper, Question 15

Probability and Statistics • Sums of Random Variables

STEP I Specimen paper, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q15

The key idea

The strange-looking distribution of C is exactly the error made when one price is rounded to the nearest 10 pence, provided the last digit of the price is equally likely to be any of $0, 1, \ldots, 9$. The shopper's total error is then a sum of 36 independent copies of C, so its mean and variance follow at once, and the Central Limit Theorem tells us how surprising a discrepancy of 20 pence is.

Solution

Expectation and variance of C

The distribution is symmetric about 0, since $\mathrm{P}(C = i) = \mathrm{P}(C = -i)$ for every i. So the terms $i\,\mathrm{P}(C = i)$ cancel in pairs, and

$$\mathrm{E}(C) = 0.$$

(The probabilities do sum to 1: $9 \cdot \frac{1}{10} + 2 \cdot \frac{1}{20} = 1$.) Since the mean is 0, the variance is just $\mathrm{E}(C^2)$:

$$\begin{aligned}\mathrm{Var}(C) = \mathrm{E}(C^2) &= 2 \cdot \frac{25}{20} + \frac{2}{10}(1^2 + 2^2 + 3^2 + 4^2) \\ &= \frac{5}{2} + \frac{2 \cdot 30}{10} = \frac{17}{2}.\end{aligned}$$

The shopper's estimate

The model. For one item, let C be the actual price minus the rounded price, in pence. This depends only on the last digit d of the price. We assume that d is equally likely to be any of $0, 1, \ldots, 9$; that is what buying "at random" should mean here, though it is an assumption about real prices.

- $d = 0$: no rounding, so $C = 0$.

- $d = 1, 2, 3, 4$: the price is rounded down by d, so $C = d$.
- $d = 6, 7, 8, 9$: the price is rounded up by $10 - d$, so $C = -(10 - d)$, which is $-4, -3, -2, -1$.
- $d = 5$: the price is rounded down or up with equal probability, so $C = 5$ or -5, each with probability $\frac{1}{2} \cdot \frac{1}{10} = \frac{1}{20}$.

Each of $-4, \ldots, 4$ arises from exactly one digit, with probability $\frac{1}{10}$, and ± 5 each have probability $\frac{1}{20}$. This is precisely the distribution in the first part, which is why it was set up that way.

The total error. Let $C_1, \ldots, C_{36}$ be the errors on the 36 items, and let

$$S = C_1 + C_2 + \cdots + C_{36}$$

be the true bill minus the shopper's estimate. Since the items are chosen at random, we take the C_j to be independent, each distributed like C. Then

$$\mathrm{E}(S) = 36 \cdot 0 = 0, \qquad \mathrm{Var}(S) = 36 \cdot \tfrac{17}{2} = 306,$$

using independence for the variance. So the standard deviation of S is $\sqrt{306} \approx 17.5$ pence.

Ask yourself Is 20 pence a large discrepancy or a small one?

It must be compared with the standard deviation, not with the size of a single rounding error. The cashier's bill is 20 pence above the estimate, which means $S = 20$. That is

$$\frac{20}{\sqrt{306}} \approx 1.14$$

standard deviations above the mean. Because S is a sum of 36 independent, identically distributed variables, the Central Limit Theorem says that S is approximately $\mathrm{N}(0, 306)$. With a continuity correction (as S takes whole-number values),

$$\mathrm{P}(S \geq 20) \approx \mathrm{P}\left(Z \geq \frac{19.5}{\sqrt{306}}\right) = \mathrm{P}(Z \geq 1.11) \approx 0.13,$$

where $Z \sim \mathrm{N}(0, 1)$. An error at least this large (in either direction) happens about 26% of the time.

So the shopper should **not** suspect a mistake. A difference of 20 pence is well within the normal spread of her rounding errors, about one standard deviation, and would happen on roughly one shopping trip in seven or eight. Only a discrepancy of more than about two standard deviations, say 35 pence or more, would give her real grounds for suspicion.

Checks

- Rounding a price to the nearest 10 pence changes it by at most 5 pence, so the worst case for 36 items is 180 pence. A typical error of about 17 pence is far smaller than this, as it should be: positive and negative errors largely cancel.
- The computer confirmed that the rounding error has exactly the distribution of C, with $\mathrm{E}(C) = 0$ and $\mathrm{Var}(C) = \frac{17}{2}$. It then convolved 36 copies exactly using fractions: $\mathrm{Var}(S) = 306$ and $\mathrm{P}(S \geq 20) \approx 0.1330$, against 0.1325 from the normal approximation and 0.132 from 2×10^5 simulated shopping trips.

Common slips

- Taking $\mathrm{Var}(S)$ as $36^2\,\mathrm{Var}(C)$. That would be the variance of $36C$, one error repeated 36 times; for independent errors the variances simply add.
- Treating the probability of S being exactly 20 (about 0.012) as the measure of surprise. Every individual value of S is fairly unlikely; what matters is the probability of an error at least as large as the one observed.
- Forgetting to say what is being assumed: that the last digits of the prices are uniformly distributed and the items' errors are independent.

Beyond STEP

This is the reasoning behind a hypothesis test: assume nothing has gone wrong (here, S is just rounding error), then ask how likely a result at least as extreme as the observed one would be. The same analysis of rounding errors is used in numerical analysis, where the errors from rounding many numbers usually grow like the square root of their count, not linearly. Blitzstein and Hwang, *Introduction to Probability*, covers sums of random variables and the Central Limit Theorem clearly.

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